

Comprehensive Airline Delay, Cancellation, and Operational Insights Analysis Using Python and Power BI

1. Project Overview:

Developed an interactive Airline Performance & Cancellation Analytics Dashboard using Power BI to analyse flight operations, delays, cancellations, airport performance, and airline efficiency. The solution enables stakeholders to monitor key performance indicators, identify delay patterns, and make informed operational decisions through data-driven insights.

Project Objective:

- Analyse airline operational performance and flight punctuality
- Track delay and cancellation metrics across airlines and airports
- Identify key factors contributing to flight disruptions
- Evaluate airline and airport efficiency using performance KPIs
- Support strategic decision-making through interactive dashboards and visual analytics
- Enhance operational efficiency and customer experience through actionable insights

2. Data Sources

• Source Description and Timeline:

https://www.kaggle.com/datasets/patrickzel/flight-delay-and-cancellation-data-2019-2023-v2/data?select=flights_sample_2m.csv

- **Domain:** Environment

3. Problem Statement

Analysing historical airline operational data from multiple carriers by cleaning, processing, and visualizing flight records using Python and Power BI. The study focuses on identifying patterns in flight delays and cancellations, evaluating airline and airport performance, understanding delay causes, and developing interactive dashboards to support data-driven decision-making and operational efficiency improvements.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
FL_DATE	Date	Date of flight operation
AIRLINE	String(text)	Name of the airline operating the flight
AIRLINE_DOT	String(text)	Airline name as per DOT records
AIRLINE_CODE	String(text)	Unique airline code
DOT_CODE	Integer (Numeric)	Department of Transportation airline identifier
FL_NUMBER	Integer (Numeric)	Unique flight number assigned by the airline
ORIGIN	String (Text)	Origin airport code
ORIGIN_CITY	String (Text)	City of departure airport
DEST	String (Text)	Destination airport code
DEST_CITY	String (Text)	City of arrival airport
CRS_DEP_TIME	Integer (Numeric)	Scheduled departure time
DEP_TIME	float (Numeric)	Actual departure time
DEP_DELAY	float (Numeric)	Delay in departure time (minutes)
TAXI_OUT	float (Numeric)	Time spent taxiing before takeoff (minutes)
WHEELS_OFF	float (Numeric)	Time when aircraft leaves the ground
WHEELS_ON	float (Numeric)	Time when aircraft touches down
TAXI_IN	float (Numeric)	Time spent taxiing after landing (minutes)
CRS_ARR_TIME	Integer (Numeric)	Scheduled arrival time
ARR_TIME	float (Numeric)	Actual arrival time
ARR_DELAY	float (Numeric)	Delay in arrival time (minutes)
CANCELLED	Integer (Numeric)	Flight cancellation status (0 = No, 1 = Yes)
CANCELLATION_CODE	String (Text)	Reason code for cancellation

DIVERTED	Integer (Numeric)	Indicates whether flight was diverted (0 = No, 1 = Yes)
CRS_ELAPSED_TIME	Integer (Numeric)	Scheduled flight duration (minutes)
ELAPSED_TIME	float (Numeric)	Actual flight duration (minutes)
AIR_TIME	float (Numeric)	Time spent in the air (minutes)
DISTANCE	Integer (Numeric)	Flight distance travelled (miles)
DELAY_DUE_CARRIER	float (Numeric)	Delay caused by airline operations
DELAY_DUE_WEATHER	float (Numeric)	Delay caused by weather conditions
DELAY_DUE_NAS	float (Numeric)	Delay caused by National Airspace System issues
DELAY_DUE_SECURITY	float (Numeric)	Delay caused by security-related issues
DELAY_DUE_LATE_AIRCRAFT	float (Numeric)	Delay caused by the late arrival of the aircraft

5. Tools & Technologies

- Python: Data cleaning, pre-processing, and analysis
- Pandas: Data manipulation and transformation
- Matplotlib: Data visualization and trend analysis
- Seaborn: Statistical data visualization
- Power BI: Interactive dashboard development

6. Data Pre-Processing (Python):

Tasks Performed:

Data Cleaning & Transformation: Removed duplicates, handled missing values, standardized formats, Splitting, Find Outliers, Find and replace

1. **Removed duplicate records to ensure each flight is counted only once.**

▶ # Removed duplicate records to ensure each flight is counted only once.
df.drop_duplicates(inplace=True)
df

2. Removed leading and trailing spaces from airport and city names

```
#Removed leading and trailing spaces from airport and city names
df['ORIGIN_CITY'] = df['ORIGIN_CITY'].str.strip()
df['ORIGIN'] = df['ORIGIN'].str.strip()
df['DEST'] = df['DEST'].str.strip()
df['DEST_CITY'] = df['DEST_CITY'].str.strip()

df
```

3. Replaced missing cancellation codes for non-cancelled flights with "Not Cancelled".

```
) #Replaced missing cancellation codes for non-cancelled flights with "Not Cancelled".
df.loc[df['CANCELLED'] == 0, 'CANCELLATION_CODE'] = (df.loc[df['CANCELLED'] == 0, 'CANCELLATION_CODE'].fillna('Not Cancelled'))

df['CANCELLATION_CODE']
```

4. Set all operational flight-related values to 0 for cancelled flights because the flight was not operated.

```
#Set all operational flight-related values to 0 for cancelled flights because the flight was not operated.

ops_cols = ['DEP_TIME', 'ARR_TIME',
            'WHEELS_OFF', 'WHEELS_ON',
            'AIR_TIME', 'TAXI_OUT', 'TAXI_IN', 'DEP_DELAY', 'ARR_DELAY', 'ELAPSED_TIME']

df.loc[df['CANCELLED'] == 1, ops_cols] = 0
df
```

[Show hidden output](#)

5. #Converted the flight date column from text format to datetime format for date-based analysis

```
#Converted the flight date column from text format to datetime format for date-based analysis.

import pandas as pd
df['FL_DATE'] = pd.to_datetime(df['FL_DATE'], dayfirst=True)

#Extracted the year from the flight date to analyze yearly trends and patterns.
df['YEAR'] = df['FL_DATE'].dt.year
```

6. Standardized departure and arrival time columns by converting them into a consistent HH:MM format.

```
#Standardized departure and arrival time columns by converting them into a consistent HH:MM format.
time_cols = ['CRS_DEP_TIME', 'DEP_TIME', 'CRS_ARR_TIME', 'ARR_TIME']

for col in time_cols:
    df[col + '_TIME'] = pd.to_datetime(
        df[col].astype('Int64').astype(str).str.zfill(4),
        format='%H%M',
        errors='coerce'
    ).dt.strftime('%H:%M')
```

7. Imputed missing values in diverted flights by replacing them with the median value from the same airline and year to maintain data consistency and reduce bias

```
#Imputed missing values in diverted flights by replacing them with the median value from the same airline and year to maintain data consistency and reduce bias.
cols = ['WHEELS_ON', 'ARR_DELAY', 'AIR_TIME', 'TAXI_IN', 'ELAPSED_TIME', 'ARR_TIME']

mask1 = (df['CANCELLED'] == 0) & (df['DIVERTED'] == 1)

for col in cols:
    df.loc[mask1, col] = (
        df.loc[mask1]
        .groupby(['AIRLINE', 'YEAR'])[col]
        .transform(lambda x: x.fillna(x.median()))
    )
```

8. Set all delay reason columns to 0 for flights that arrived on time or early, since no arrival delay occurred.

```
#Set all delay reason columns to 0 for flights that arrived on time or early, since no arrival delay occurred.
delay_cols = ['DELAY_DUE_CARRIER', 'DELAY_DUE_WEATHER', 'DELAY_DUE_NAS', 'DELAY_DUE_SECURITY', 'DELAY_DUE_LATE_AIRCRAFT']
df.loc[df['ARR_DELAY'] <= 0, delay_cols] = 0
```

9. Removed extreme outliers from numerical columns using the IQR method and compared the dataset size before and after cleaning

```
#Removed extreme outliers from numerical columns using the IQR method and compared the dataset size before and after cleaning.
```

```
for col in num_cols:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.8 * IQR
    upper = Q3 + 1.8 * IQR

    df = df[
        (df[col] >= lower) &
        (df[col] <= upper)
    ]

print("Original Shape:", df_clean.shape)
print("New Shape:", df.shape)
print("Rows Removed:", len(df_clean) - len(df))
```

```
Original Shape: (100000, 38)
New Shape: (73604, 38)
Rows Removed: 26396
```

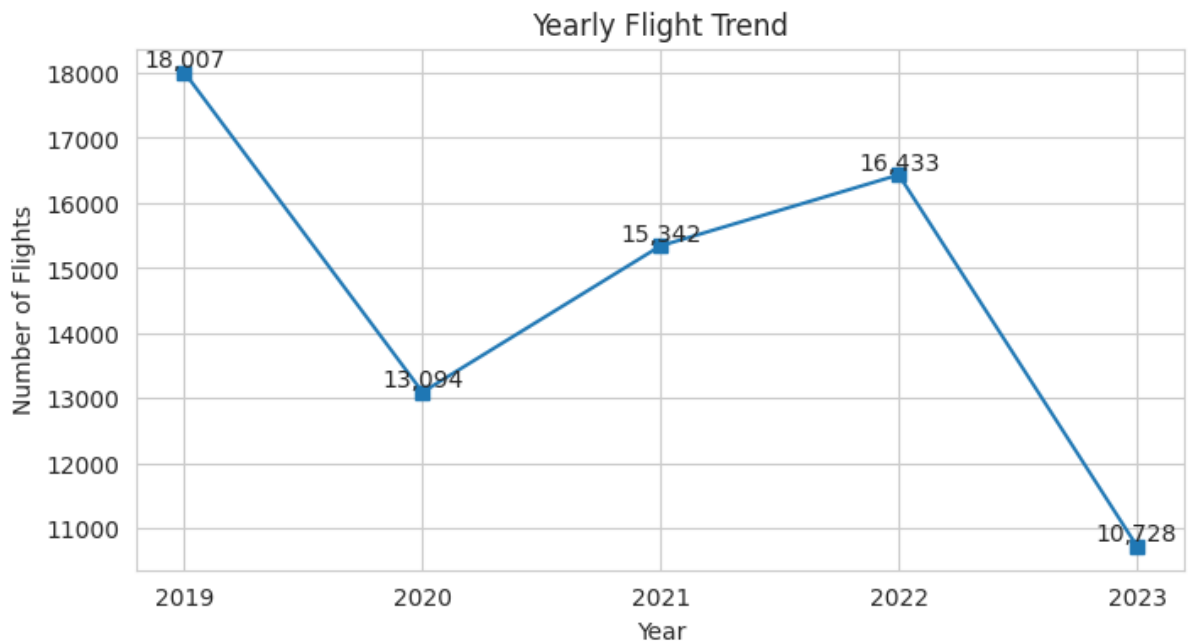
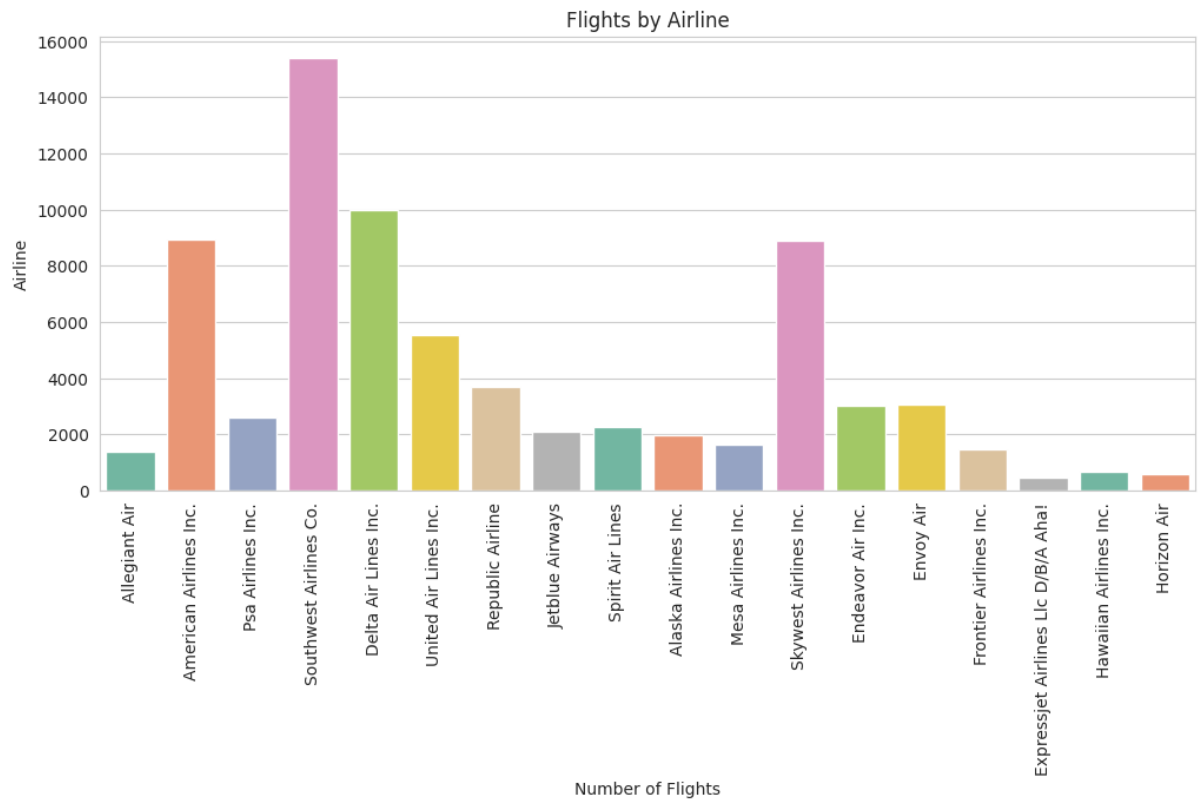
10. find shape After removed outliers

```
#find shape After removed outliers
df.shape
```

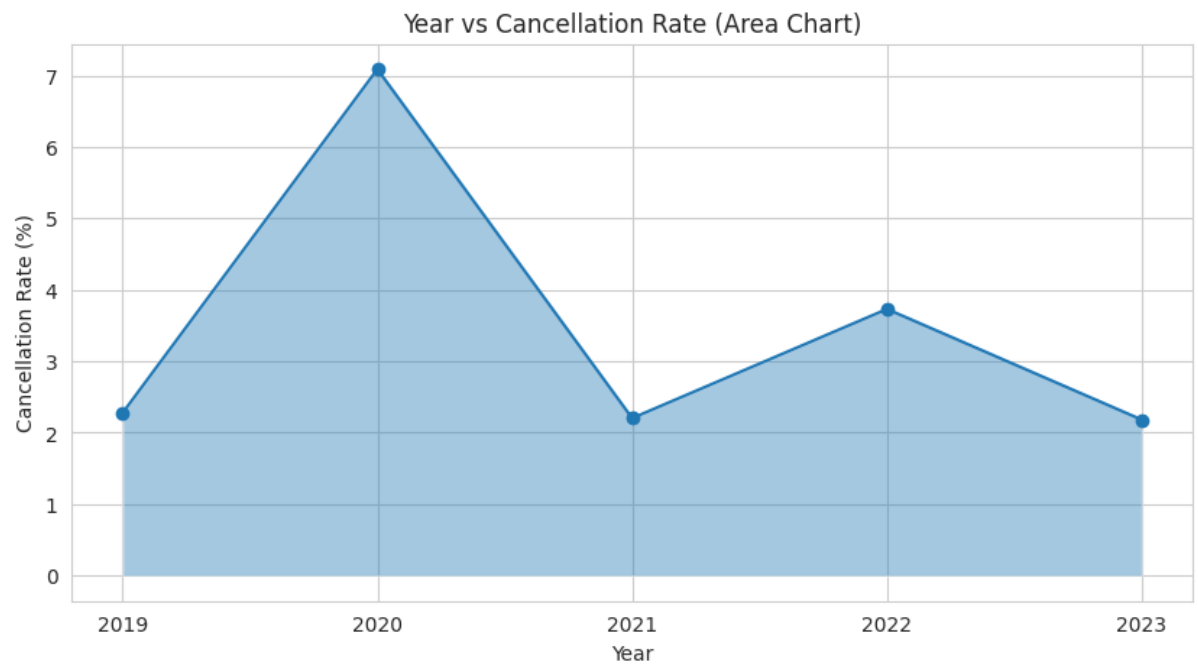
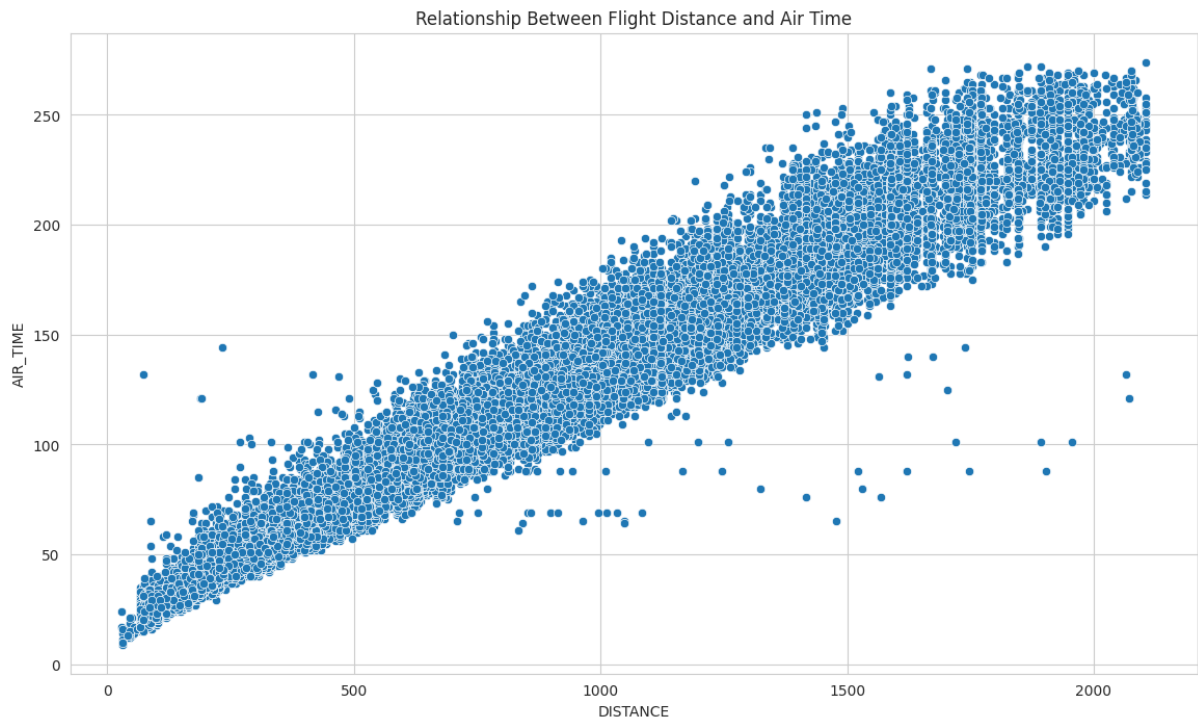
```
(73604, 38)
```

11. EDA and Visualizations

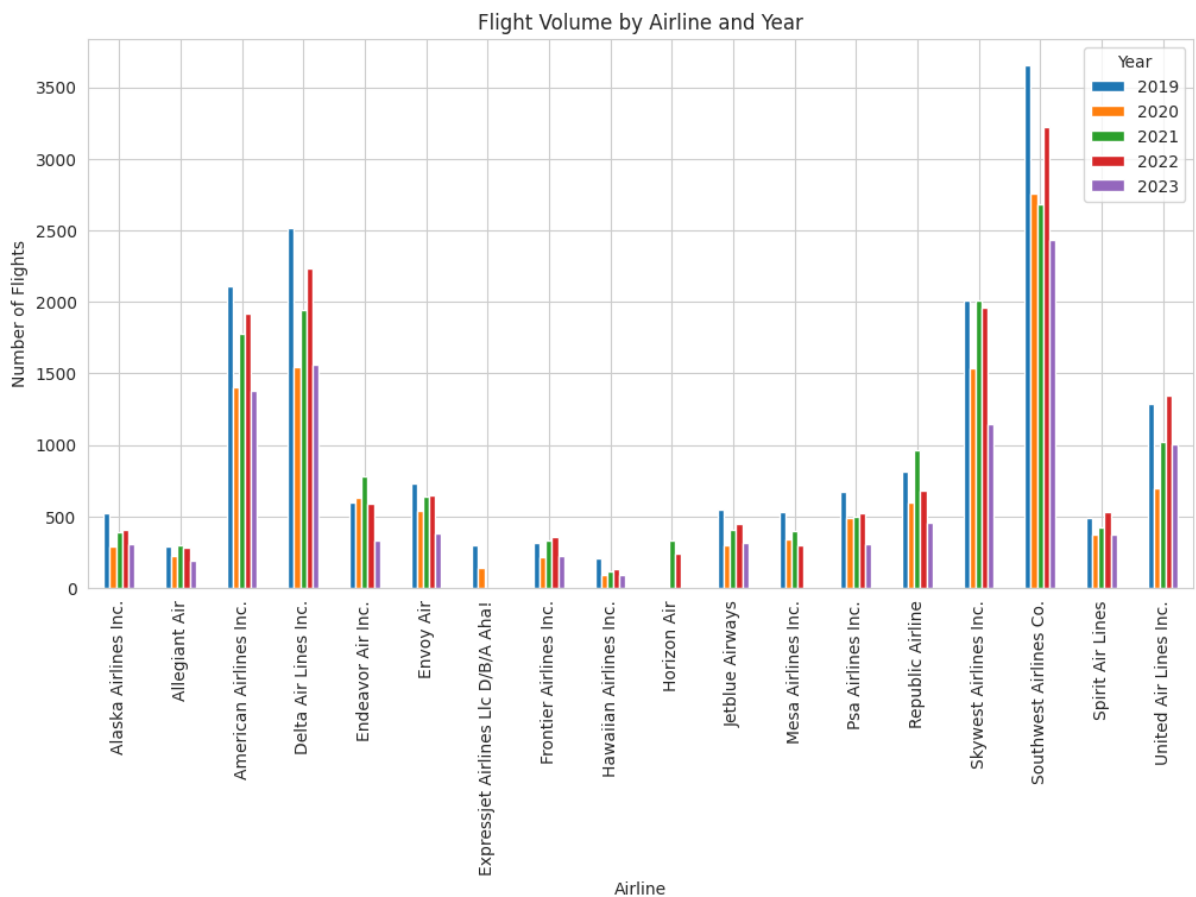
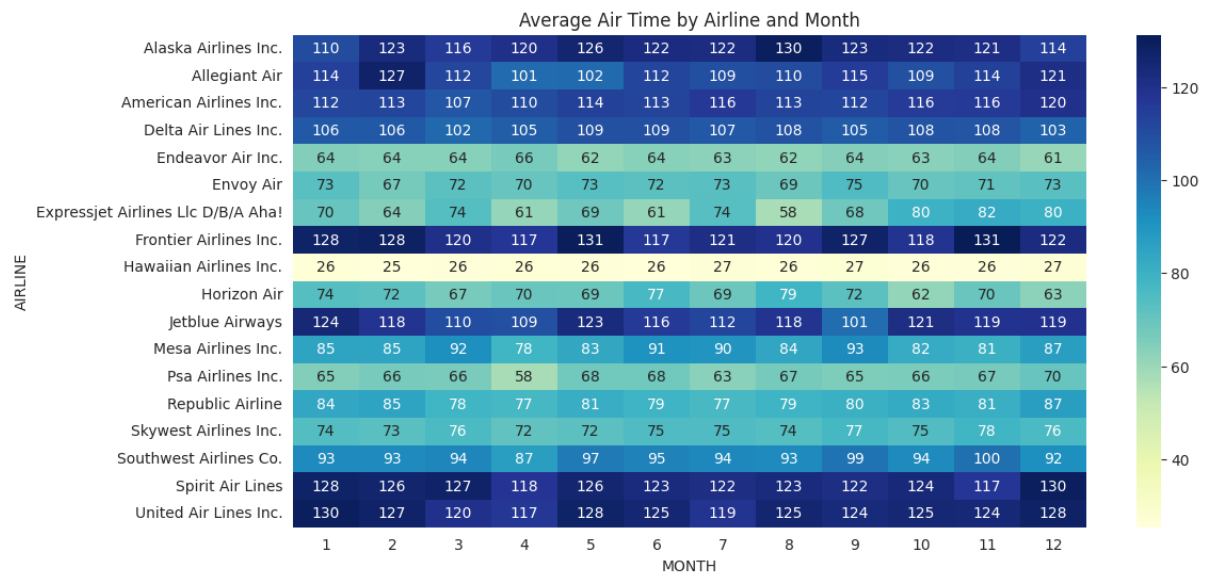
1. UNIVARIATE ANALYSIS



2. BIVARIATE ANALYSIS



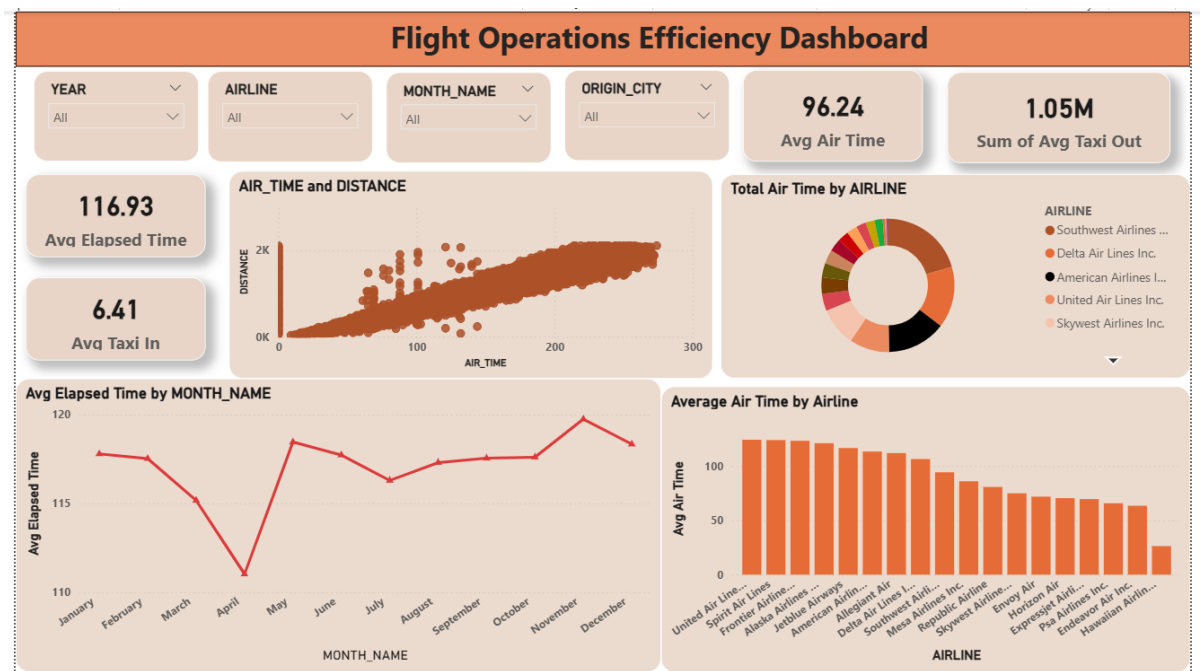
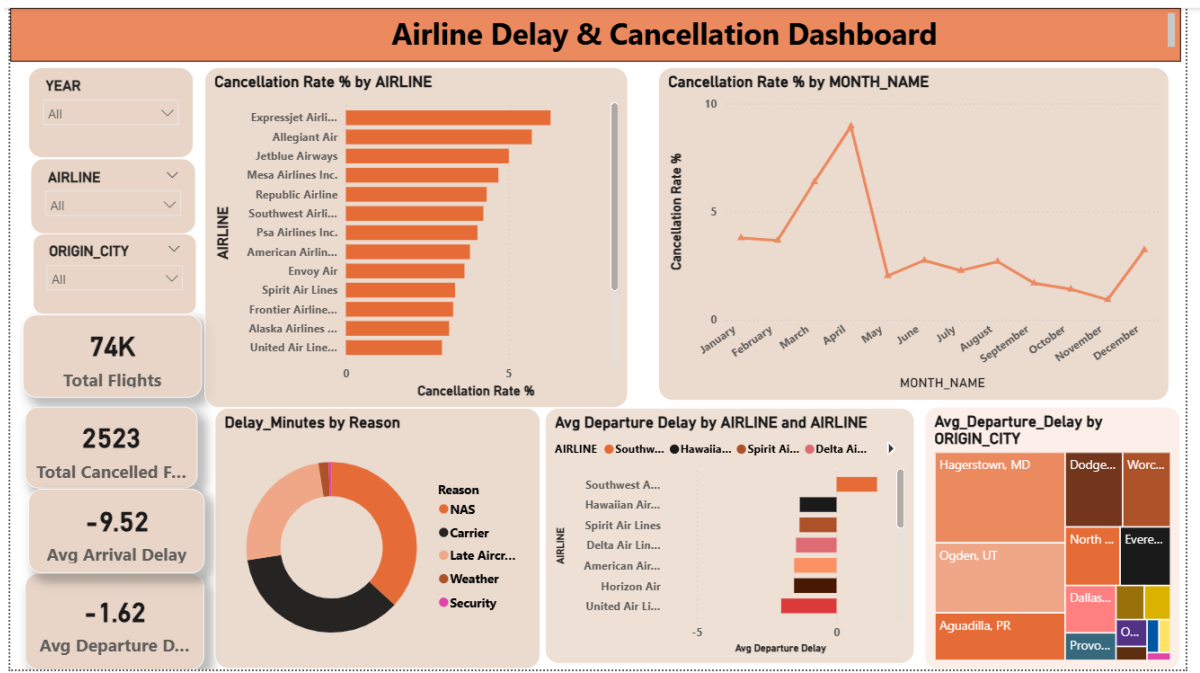
3. MULTIVARIATE ANALYSIS



7.Calculated DAX Measures:

Implemented DAX formulas for key metrics, such as Total Flights, Total Cancelled Flights, Average Departure Delay, Average Arrival Delay, Average Taxi in, Average Elapsed Time and Average Air Time

8. Analysis and Visualizations (Power BI) Dashboard Features:



9. Insights & Conclusion:

Provide the analysis insights:

Descriptive

- Total **flight records were successfully analysed** from the cleaned airline dataset
- A significant number of flights experienced **delays and cancellations**, indicating operational challenges.
- Certain airlines recorded **higher cancellation rates**, while others maintained relatively stable performance.
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- **NAS (National Aviation System)** was identified as the **major contributor to delays**, followed by carrier and late aircraft issues.
- Monthly trends show that **cancellations and delays vary seasonally**, with some months experiencing higher disruptions.
- The **average arrival and departure delays** indicate that flights are sometimes **early or slightly delayed**, showing schedule adjustments.
- Some **origin cities recorded higher delays**, suggesting location-based operational inefficiencies.
- • A strong relationship exists between **air time and distance**, indicating consistent flight behavior.
- **Taxi-out time contributes significantly** to total flight time, reflecting airport congestion.

Diagnostic

- Airlines with higher cancellation rates may be impacted by:
 - Poor operational planning
 - Fleet or maintenance issues
 - Route-specific challenges
- High NAS delays indicate:
 - Air traffic congestion
 - Airport system inefficiencies
 - Weather-related disruptions

- Seasonal spikes in delays and cancellations may be due to:
 - Increased passenger demand
 - Weather conditions (storms, winter effects)
- Higher taxi-out times suggest:
 - Runway congestion
 - Inefficient airport ground operations
- Variations across cities indicate:
 - Infrastructure limitations
 - High traffic airports
 - Regional weather patterns
- Differences in airline performance highlight:
 - Variation in operational efficiency
 - Route planning and scheduling strategies

Predictive

- Airlines with consistently high delays and cancellations are likely to:
 - Continue facing operational inefficiencies
 - Experience lower customer satisfaction
- Peak travel seasons are expected to:
 - Increase delay and cancellation rates
- If NAS-related issues persist:
 - System-wide delays may continue to rise
- Airports with high congestion may:
 - Experience increased taxi times and longer delays
- Airlines with optimized operations will:
 - Gain competitive advantage
 - Improve customer trust

- Flight duration and delays may increase if:
 - Traffic volume grows without infrastructure improvements

Prescriptive

- Improve air traffic management systems to reduce NAS delays
- Optimize flight scheduling and route planning
- Reduce taxi-out time by improving runway and ground operations
- Increase operational focus during peak months and high-risk seasons
- Enhance airport infrastructure in high-delay cities
- Implement predictive analytics models to forecast delays and cancellations
- Improve airline maintenance and fleet management **systems**
- Provide **real-time updates and communication to passengers**
- Use **data-driven decision-making tools (Power BI, AI models)** for continuous monitoring

10.Conclusion

- Successfully analysed airline operational data using Power BI
- Identified key trends in **delays, cancellations, and performance**
- Highlighted **monthly variations and high-risk periods**
- Detected major causes of delays like **NAS, carrier, and congestion**
- Provided insights into **air time, taxi time, and route efficiency**
- Helps improve:
 - Operational efficiency
 - Resource utilization
 - Flight scheduling accuracy
 - Passenger experience
- Enables **data-driven decision-making** to reduce disruptions
- Supports better planning through **trend and pattern analysis**

